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# Models of symbol emergence in communication: a conceptual review and a guide for avoiding local minima

**Julian Zubek[1] · Tomasz Korbak[2] · Joanna Rączaszek-Leonardi[1]**

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**Abstract**
Computational simulations are a popular method for testing hypotheses about the emergence of symbolic communication. This kind of research is performed in a variety of traditions including language evolution, developmental psychology, cognitive science, artificial intelligence, and robotics. The motivations for the models are different, but the operationalisations and methods used are often similar. We identify the assumptions and explanatory targets of the most representative models and summarise the known results. We claim that some of the assumptions—such as portraying meaning in terms of mapping, focusing on the descriptive function of communication, and modelling signals with amodal tokens—may hinder the success of modelling. Relaxing these assumptions and foregrounding the interactions of embodied and situated agents allows one to systematise the multiplicity of pressures under which symbolic systems evolve. In line with this perspective, we sketch the road towards modelling the emergence of meaningful symbolic communication, where symbols are simultaneously grounded in action and perception and form an abstract system.

**Keywords** Emergent communication · Signalling games · Symbol grounding · Compositionality · Computational simulations · Linguistics · Ecological psychology

✉ Julian Zubek
j.zubek@uw.edu.pl

[1] Human Interactivity and Language Lab, Faculty of Psychology, University of Warsaw, Stawki 5/7, 00-183 Warsaw, Poland

[2] Department of Informatics, University of Sussex, Brighton, United Kingdom

## 1 Introduction

Symbolic nature is commonly thought to demarcate human communication from communication in most other species (Deacon 1997; Beecher 2021).[1] Human languages exhibit unparalleled flexibility and constitute the basis for other symbolic systems crucial for modern human cognition (e.g. formal systems such as mathematical notation). Thus, the problem of the emergence of symbolic communication systems with formal properties from continuous interactions in a continuous (analog) world is central to theories of cognition. Over the last three decades, this problem has been tackled using computational models, which has made the theories more internally coherent and more comparable and classifiable. Most communication emergence models involve simulated agents that exchange messages over a communication channel while acting in some environment (Ohmer and Cuskley 2023). These agents are typically rewarded for successfully achieving specific goals related to communication, leading to the development of efficient communication protocols.

In the present paper, we offer a critical review of agent-based communication emergence models relevant to the symbol emergence problem. The overall logic of the paper is as follows: we recognise that models are grounded in different domains of inquiry—from philosophy to linguistics to robotics—and motivated by questions pertinent to those domains (Sect. 2). We subsequently describe several concrete models representative of the multitude of works that have appeared over the last three decades (Sect. 3). The scope of what is being modelled ranges from highly abstract models rooted in evolutionary game theory to embodied simulations with agents navigating complex environments. We evaluate the models in terms of their merits and shortcomings, which helps us to pinpoint their underlying common assumptions about the communication process and to identify those that hinder progress towards modelling the emergence of structured and contextually-flexible symbolic systems (Sect. 4). We identify three such assumptions:

1. Taking the existence of discrete symbols for granted.
2. Assuming that the referring or world-descriptive function is the main function of communication.
3. Adopting the mapping metaphor as the relation between symbolic forms and their meanings.As we claim in this paper, the communication emergence field is in a local minimum created by this particular set of assumptions underlying extant modelling efforts. We argue that certain properties of the communication process are currently taken for granted, while crucial aspects of language remain unexplored. The paper's main claim is that progress does not hinge on complexifying agents' internal architectures for computational efficiency, enriching tasks or environments, nor creatively composing models. Instead, to escape the local minimum, we must closely examine and potentially suspend the three core assumptions. We propose relaxing these assumptions while recognising multiple sources of communication complexity arising from individual, social, cultural, and evolutionary pressures (Sect. 5). Only by combining

[1] There are arguments that animals such as domestic dogs (Olney 2013) or chimpanzees (Cissewski and Luncz 2021) may learn to understand and use symbolic references to some extent. Deacon (1997) believes that it is possible with a special training procedure, but in principle does not occur in animal communication systems found in nature. At the same time, he recognises that animal communication is continuous with human communication with respect to other types of signs (icons and indices).

these sources can we avoid a local minimum stemming from a one-dimensional treatment of communication as an optimisation process with a single goal. We demonstrate that suspending the aforementioned assumptions and recognising the emergent nature of signals and symbols within coordinative dynamics facilitates identifying sources of complexity of linguistic structures, the interactions of these sources, and their relative contributions to symbol emergence. These postulates are realised by a certain strain of embodied simulations (Sect. 6). By opening this possibility, we hope to establish novel explanatory targets for models and new avenues for research on emergent symbolic communication.

Before we proceed, we will discuss some historical influences that have shaped current communication emergence models. The first influence concerns the notion of symbol itself and its evolution over time. As Deacon (2011b) observed, historically the term "symbol" was applied almost exclusively to language utterances, inscriptions, graphical markings, etc., created specifically to represent something. The overarching idea was that symbols are created culturally according to some convention. Congruently, in Peircean semiotics, symbols are a particular class of signs (along with icons and indices) whose meanings are determined by referring to a general law (Peirce 1903). The law has to be applicable to a wider class of signs, not just to single instances (symbols are general), and the person interpreting a symbolic sign has to assume that the originator of the sign adheres to this law (symbols are conventional).[2] For example, a specific word belongs to a grammatical category (a general class), and we presume that a user of this word comprehends grammatical conventions rather than simply reproducing memorised phrases. This is the notion of symbol we have in mind when we write about symbol emergence in this paper. However, throughout the 20th century, emphasis shifted, and the concepts of generality and conventionality of symbols became somewhat conflated. The distinguishing feature of symbols became the arbitrariness of the connection between symbol and its referent (Deacon 2011b). Then, in some technical disciplines, symbols began to be narrowly understood as discrete tokens that can be manipulated formally and constitute parts of a code—a mapping between symbols from two distinct domains. As we will argue later, existing models of emergent communication use the term "symbol" in various ways but demonstrate a tendency to slip into this last, technical understanding. This tendency is discussed extensively by Taniguchi et al. (2019), consistently with our views.

The second influence on communication emergence modelling is the view of communication imposed by the influential work of Shannon in the mid-20th century and strengthened by the facilitation of computational modelling that such a perspective carries. Most models portray communication as a method of conveying messages from an information source to a destination via signals transmitted over a communication channel (Shannon 1948). In Shannon's words, "The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point" (ibid., p. 378). This highlights the problem of message transmission as central to communication. For a long time, the "meanings" of messages were not emphasised and were usually simplified as a mapping between symbols and objects in the agent environment, which realises the

[2] Peircean categories are notoriously abstract and challenging to interpret, compounded by Peirce's development of these concepts over multiple decades across various manuscripts. Consequently, multiple interpretations of Peirce's semiotics exist; for context, see Liszka et al. (2006).

"grounding" of symbols (congruent with the narrow, technical notion of symbol that we describe above) (Harnad 1990). The idea of a message-object mapping, which represents meaning, mirrors the idea of a message-signal mapping central to information theory (where a "signal" is an element of the code used for transmission and a "message" is what is transmitted with the code).

However, the fathers of information theory were convinced that the term "communication" should have a broader meaning. As Weaver (1949) put it, "The word communication will be used here in a very broad sense to include all of the procedures by which one mind may affect another" (ibid., p. 10). Weaver's definition resonates with pragmatic accounts of meaning that focus on influencing agent actions rather than representing the world. In human communication, we often use existing conventions to effectively influence another person. For example, I can signal a person to come closer or stay away using conventional gestures. Conventions are central to the broader notion of symbol present in semiotics. We find a formal treatment of conventions in Lewis' signalling game (from now on the LSG; Lewis (1969))–the third historical source of assumptions concerning the communication process.

Based on game theory, Lewis demonstrated that a conventional communication protocol emerges as an equilibrium when two agents—sender and receiver—are rewarded for successful coordination; it is rational for the two agents to stick to the convention, even if the convention itself is arbitrary. This was Lewis' solution to the problem of meaning creation by convention. The LSG can be formally defined as a game between a sender $s : \mathcal{I} \to \mathcal{M}$, who sends a message $m \in \mathcal{M}$ upon observing an input $i \in \mathcal{I}$, and a receiver $r : \mathcal{M} \to \mathcal{A}$, who takes an action $a \in \mathcal{A}$ upon receiving a message $m \in \mathcal{M}$. A payoff is given to both the sender and the receiver depending on the sender's input and the receiver's action.

Notably, in the original formulation of the LSG, information-theoretical tools were not used, and Shannon's communication theory was not cited as a reference. Later, the two approaches were integrated by Skyrms (2010), who subsumed the LSG into the general information-theoretical schema for communication. He interpreted signals as probabilistic events that carry quantifiable information and simultaneously perform coordinative function. However, this reinterpretation undermine some of the pragmatic inclinations of Lewis' original work. Skyrms noted that in many situations the LSG can be discussed equivalently in terms of information flow and coordination, but he preferred the first perspective, which reversed Lewis' original emphasis.[3] This made it easy to once more conflate signalling with information transmission and to see the LSG as part of the meaning-mapping enterprise.

The resulting framework, realised by many computational models, can be illustrated by the example used by Wittgenstein (Wittgenstein 1953; Correia 2020), where a builder uses building stones of various shapes and asks his assistant to pass a stone of a particular shape. These kinds of structured interactions are simulated with artificial agents, whose behaviour is governed by internal adaptive mechanisms (for example, weight updates in neural networks). At first, agents choose from their predetermined sets of behaviours and signals at random. Each episode of the game has consequences for the agents (for instance, passing a right or wrong stone), which determines their reward, and, in turn, makes them update both their behavioural and signalling predispositions. In this way, signals become conventions in

[3] In Lewis (1969), the word "information" appears 11 times, and the word "coordination" appears 138 times; in Skyrms (2010), the word "information" appears 358 times, and the word "coordination" appears seven times.

Lewis' sense, resulting in the individual coupling of signals to agents' behaviours (policies) and the sharing of this coupling among agents. This adaptation process is thus a coevolution within a population rather than a simple optimisation of individual fitness. Computational simulations demonstrate how such iterative adaptation can lead to the emergence of stable communication systems.

Existing work on emergent communication has repeatedly demonstrated the possibility of a conventionalised mapping from a set of discrete signals to a set of objects emerging without any central authority. Optimal mapping can be learnt in various settings and with different agent architectures and adaptation mechanisms (Spike et al. 2017). However, natural languages, owing to their richness and expressiveness, do not resemble simple bidirectional mappings. They include parts of speech that are not easily mapped to objects or events, they involve complex syntax and grammar, they exploit ubiquitous nuances and ambiguities, they allow speculative conversations on abstract entities, and they crucially depend on interaction in time. These properties are visible in everyday verbal interactions, which are full of ambiguity and often involve repair sequences (imagine asking the clerk at a store for a certain object and receiving a puzzled look followed by a clarification question) (Dingemanse et al. 2015).

Much work is still needed to build an adequate account of these phenomena within the field of language emergence modelling. This is not to say that efforts are not being made to develop the field and understand the complexity of the problem of the emergence of symbolic communication—for reviews see Hutchins and Johnson (2009); Lazaridou and Baroni (2020); Lipowska and Lipowski (2022); Ohmer and Cuskley (2023); Brandizzi (2023); Galke and Raviv (2024); Boldt and Mortensen (2024); Peters et al. (2024). The present review differs from the previous work that aimed to summarise modelling efforts in that it strives to identify and examine the very foundations of the assumed phenomenon of symbolic communication, which most works seem to take for granted. Previous reviews were very helpful in identifying the missing sources of symbolic structure, which were revealed by gradually including aspects of developmental and interactive contexts of communication, such as learning bottlenecks, task structuring, kinds of collaboration, and the richness and complexity of the perceptual information. Nevertheless, the overall "skeletal dynamics" of the models remain similar (Ohmer and Cuskley 2023) and, as we claim above, rooted in the information-theoretic interpretation of signalling games.

One of the most conceptually advanced early reviews of the models of communication emergence was presented by Hutchins and Johnson (2009). They rightly point out that models rarely address the heart of the problem, because they are explicitly seeking a mapping between predefined symbols and predefined meanings ("references"). Only some models allow for the emergence of meanings in interactions (e.g. Steels and Belpaeme 2005) and only a small proportion of these seek to show the true emergence, which requires that both the meanings and the signalling systems emerge in interactions from scratch (Marocco and Nolfi 2007; Hazlehurst and Hutchins 1998; Lieck and Rohrmeier 2021).

Hutchins and Johnson advocate the latter approach and show that the problem of signal emergence involves the problem of distinguishing signalling behaviours from the ongoing flow of interaction. They illustrate this with an analysis of signal emergence in chimpanzees interaction, where infants begin to reenact frozen fragments of movement trajectories that appeared in previous interactions as communicative "gestures." The authors observe a "remove between forms and meaning" as these gestures diverge from the original trajecto-

ries. They outline various dimensions of this divergence, and suggest how these distinctions could be implemented in computational models of communication emergence. However, the relation of mapping itself (between the forms and meanings) is not questioned even in this very deep conceptual analysis.

Remaining in the aforementioned local minimum, which involves a mapping metaphor, hinders efforts to scale up from simple games to structured communication systems utilised by agents for flexibly coordinating dynamic interactions within their environments. Reviewing existing models leads us to the critical claim that a deep rethinking of foundational assumptions is necessary. It also leads to a constructive claim that recent developments in cognitive science, linguistics, and linguistic anthropology, which emphasise the primacy of action and co-action for the theory of cognition and language (Di Paolo et al. 2018), foreground interaction (Dingemanse et al. 2023) and coordinative dynamics (Tognoli et al. 2020), allow for a reformulation of the communication process itself. These developments also provide opportunities for scaling up. They enable communication to be grounded anew in biological information processes (Pattee 1969, 1982; Deacon 2011a; Rączaszek-Leonardi and Kelso 2008) and symbolic communication to be viewed as interaction-dominant and pragmatics-oriented (Kempson et al. 2000; Rączaszek-Leonardi 2016; Gregoromichelaki and Kempson 2019). Making this shift of perspective could help overcome the prevalent tendency to perceive communication as relying on mapping between amodal signs[4] and bound meanings, primarily geared towards world description.

## 2 Modelling traditions and explanatory targets

Computational models of the emergence of communication come from various research traditions, reflecting their theoretical goals, definitions, and methods (e.g. Hutchins and Johnson 2009; Lazaridou and Baroni 2020; Lipowska and Lipowski 2022; Ohmer and Cuskley 2023; Galke and Raviv 2024). No single publication can do justice to their recent proliferation. We roughly group them by discipline to draw the reader's attention to different explanatory roles assigned to simulations. These arise from differences in the perspectives on language in general and the place, role, and nature of language within human cognition and behaviour. We selected models from various disciplines to showcase the diversity of explanatory targets across research areas. The (admittedly simplified) picture in Fig. 1 illustrates some of the commitments and explanatory targets of the models in different scientific disciplines (the division of the domains is an idealisation). As illustrated, some explanatory targets are shared by multiple disciplines.

Philosophical reflection on the nature of language and its relation to thought is probably as old as philosophy itself (Seuren 1998). The goal of *naturalising truth and meaning* is to show how these concepts can be understood as natural phenomena without resorting to metaphysics that allow immaterial ideas to exist separately from physical reality. In this context, game-theoretic models have been used in research on social conventions, normativity, and semantic information,[5] starting with the influential work of Lewis (1969).

[4] We use the term "amodal" in this paper to mean "devoid of form." Amodal signs lack physical or perceptual properties.

[5] We use the term "semantic information" to distinguish between information with semantic meaning and, for instance, purely correlational information as a quantity in information theory. See Floridi (2010) for a

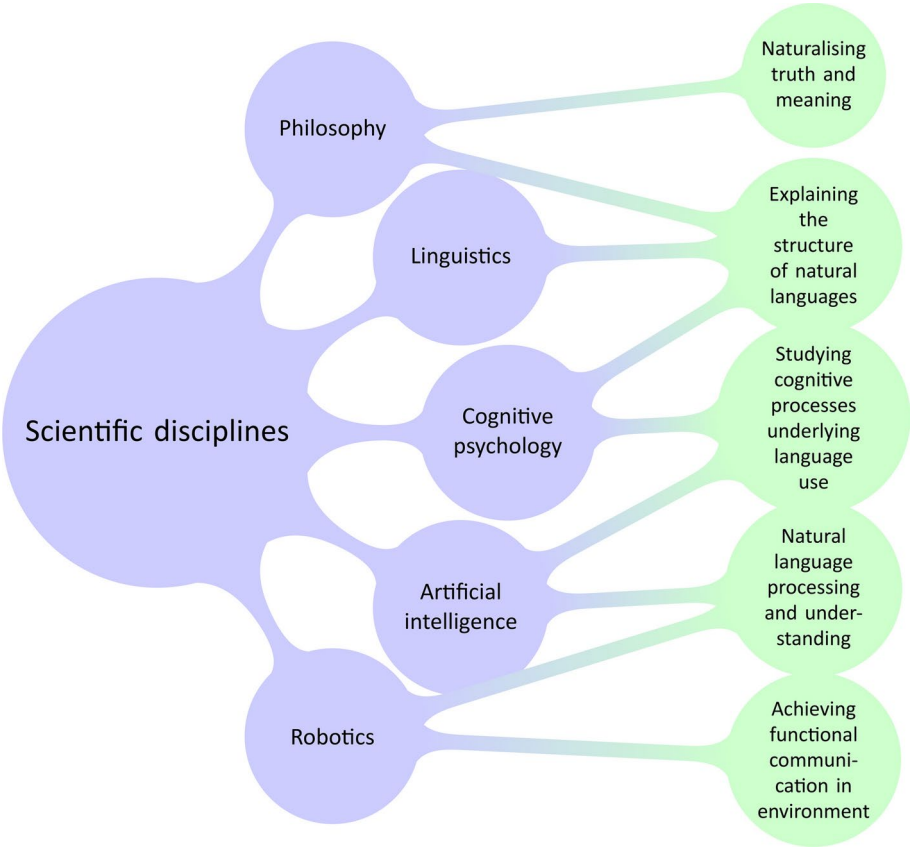


**Fig. 1** Different scientific disciplines and the roles they assign to emergent communication models

He formalised the notion of convention, which arises in the interaction of agents. In his formalism, the sender has to communicate the state of the world to the receiver by emitting discrete signals, upon which the receiver produces a reaction. If the receiver's reaction is adequate given the state of the world, this is beneficial for both agents. Mutual benefit make the agents stick to the established convention of linking signals with states of the world, even if those links are arbitrary.

Lewis' original motivation was to provide a game-theoretic explication of the notion of truth by convention (solving Quine's (1936) problem). His ideas influenced many other authors, who subsequently studied this formalism using computer simulations. The simplest implementation of the LSG consists of disembodied agents passing discrete tokens. No structured signals or environmental constraints are simulated. The goal is not to faithfully portray actual communication processes but rather to support certain philosophical positions by demonstrating the theoretical possibility of the bottom-up emergence of conventions. In recent years, this thread has been developed to account for more complex communication processes (Skyrms 2010; Barrett and Skyrms 2017; Barrett et al. 2019; Korbak et al. 2021; Ohmer and Cuskley 2023).

discussion of this term.

In linguistics, computational models of language emergence are used to ask various questions. *Explaining the structure of natural languages* means a) describing the regularities or "rules" that natural languages follow and b) explaining how these regularities emerge in the process of language evolution. Unlike biological evolution, language as such (before writing) leaves no "fossils", so the communicative systems of early humans (or hominids) are difficult to uncover. Therefore, agent-based simulations became attractive tools for testing hypotheses about the origins of language and possible evolutionary trajectories, both on biological and cultural time scales (Smith et al. 2003). The goals are theoretical: models concern various aspects of language, such as the evolution of lexemes and systematic variation of syntax or vocabulary (Nowak and Krakauer (1999); for a review, see, e.g., Ferri et al. (2018); Grifoni et al. (2016)). Compared to the philosophical tradition, simulations developed within evolutionary linguistics tend to be more focused on language structure. The simulation results are (often) compared with the empirical data to assess their plausibility as models of language evolution and change (Brighton et al. 2005a, b; Parisi et al. 2008; De Boer and Zuidema 2010).

Cognitive psychologists are also interested in *studying cognitive processes underlying language use* (language learning, understanding and production). They ask questions about what cognitive capabilities are required to effectively learn and use language and how language influences other cognitive processes, such as memory, categorisation, and reasoning. These questions can be answered with the help of computational models of language emergence or language learning that elucidate in detail the cognitive structures of an individual agent.

For example, Elman (1993) explored the relation between language learning and memory capacity, showing that a simple recurrent neural network successfully learns the structure of complex sentences only if it starts with a limited memory and only gradually "matures" to its full capabilities. A prominent group of models explored the structure of perceptual categories of an individual as shaped by a communication system emerging within a population (Steels and Belpaeme 2005; Baronchelli et al. 2015; Steels 2015). Then, the interest in cognitive science shifted from purely individualistic processes to more interactive phenomena (Deacon 1997; Brighton et al. 2005a, b; Rączaszek-Leonardi 2009). As a result, some models explored interactive factors within social environments or various demographic characteristics, such as population size and social structure, on language evolution (Vogt 2009; Dale and Lupyan 2012; Zubek et al. 2017).

Another problem studied by cognitive psychology is seeking the sources, reasons, and mechanisms of discretisation, that is, perceiving continuous spaces through discrete categories. We view this as a subproblem of categorical perception. In the context of communication emergence, discretisation can occur in both the domains of forms (signals used for communication) and meanings (concepts being communicated). At the level of forms, the self-organisation of the vowel system under communicative pressures was proposed by Lindblom et al. (1984) and de Boer (2001), while Oudeyer (2005) modelled the emergence of discrete vocalisations. At the level of semantics, a conceptual model was proposed by Gärdenfors, where discrete concepts were represented as convex regions in continuous conceptual spaces (Gärdenfors 2004). Feldman (2012) discussed the circumstances under which such representations are viable. A principled abstract model of form discretisation in a form-meaning mapping within communication was presented by Lieck and Rohrmeier (2021). In this model, forms and meanings originated in different spaces, and simulations

showed several conditions that may lead to form discretisation in signal transmission scenarios. Similar processes were also explored by Geffen Lan et al. (2020). Such discretisation processes can be seen as part of the problem of symbolic emergence, where forms are perceived as different instances of the same token and distinct from other tokens. Taniguchi et al. (2023) proposed that the simultaneous alignment of discrete categories and negotiation of their names in a population of agents can be treated as a single probabilistic generative model, making it a case of distributed Bayesian inference. This is an argument for viewing the emergence of discrete categories in communication systems as a collective-level phenomenon rather than an individualistic cognitive process (Taniguchi 2024).

Emergent communication models have also experienced a recent surge in interest in the machine learning and artificial intelligence (AI) community. Initial goals of the AI programme included replicating the cognitive processes necessary to develop human-level intelligence and use natural language the way humans do. Nowadays the field has more pragmatic orientation, and focuses on developing artificial agents capable of flexible and goal-directed language use. Recently, there has been a lot of progress in natural language processing (NLP) using semi-supervised models using large quantities of existing data (such as large language models). There are concerns that language learned in this way is not properly grounded and does not integrate well with agent's actions in the world (Mordatch and Abbeel 2018). Chat-based large language models fine-tuned using human feedback and reinforcement learning provide some pragmatic grounding; there are also some works that attempt to use multi-modal corpora of text aligned with images to visually ground language understanding (e.g. Peng et al. 2023; Liu et al. 2023). Harnad (2024) discusses conceptual limitations of these approaches, as they still attempt to ground the language "top-down" starting from huge volumes of data instead of agent's sensorimotor experience.

In contrast to data-driven NLP models, emergent communication models involve simulated agents engaging in interactions in order to evolve their own communication protocols in a bottom-up manner. Much research has attempted to replicate the basic structure of natural languages, often targeting language compositionality. Studying compositionality aligns with the goal of *studying cognitive processes underlying language use*, although the AI community, in comparison to cognitive psychology community, puts a greater emphasis on the practical results of the simulations rather than their theoretical underpinnings. Lazaridou et al. (2018) demonstrated that agents trained with reinforcement learning can solve the LSG in which input is given as raw pixel images, not symbolic tokens. Emerged protocols have displayed some degree of a systematic structure, but they were not perfectly compositional. Moreover, these protocols were quite fragile, as small changes in input stimuli or vocabulary size led to different organisation of the communication. Several other studies focused on the circumstances under which a more structured language appears (Mordatch and Abbeel 2018; Andreas 2019; Chaabouni et al. 2020).

Some other models originating in the AI community are even more directly motivated by *natural language processing and understanding*. The work of Kottur et al. (2017) on emergent dialogue is to some extent motivated by the work on visually-grounded conversational systems (Mostafazadeh et al. 2017). The formulation of the problem in Lazaridou et al. (2017) is motivated by image captioning (Karpathy and Fei-Fei 2017). Lee et al. (2018) proposed using the LSG as a blueprint to train a system translating between two natural languages without parallel corpora and instead using visual input to ground meanings. The potential applications of emergent communication research in building explainable AI

(Doshi-Velez and Kim 2017) are another active area of research. The idea is to translate a communication protocol emerging between agents into a natural language such as English. For visually grounded approaches, this can be done using labelled images by relying on visual semantics determined by the class hierarchy of the images (Lazaridou et al. 2017). Another approach is to align English utterances in an emergent communication protocol with respect to the effects they exert on a trained receiver (Lazaridou et al. 2020).

The trend of making simulations more realistic and embodied has been taken even further in artificial life and cognitive robotics communities. In this tradition, agents are autonomous mobile robots (usually simulated in silico to some extent). This means their interactions with the environment are usually much richer than in other traditions. These interactions are given central focus and communication emergence is treated as another layer of the intricate web of interactions. The goal is to *achieve functional communication in an environment*, and this environment should ultimately be an actual physical environment, not just a simulation.

To some degree, this goal could be achieved by teaching agents a preexisting language, but many researchers are also interested in how an efficient language specifically tailored to a given environment can evolve on its own. In such models, agents' communication is intertwined with the rest of their activities, often involving exploring the environment and searching for food or other agents (Floreano et al. 2007; Grouchy et al. 2016; Schulz et al. 2012). Stress is placed primarily on the robustness of emerging communication protocols and minimal starting assumptions, as communication should scale up to realistic environments. Some works in this tradition have aimed to explain basic aspects of animal communication in different evolutionary scenarios (Floreano et al. 2007; Mitri et al. 2009), while others have targeted phenomena characteristic of human languages, such as its productivity, presence of grammatical structures, and the focal role of dialogue (Grouchy et al. 2016; Schulz et al. 2012).

In contrast to other domains, signals appearing in robotics models are often modulated by multiple factors and are not fully controlled by agents. For example, in Floreano et al. (2007), communication was realised through light signals whose colours were regulated by the sender and properties that depended on the physics of the environment (e.g. directionality, the possibility of being obscured by obstacles). This level of complexity emerging from non-trivial embodiment sets this tradition apart from purely computational AI modelling, which typically relies on simpler simulated environments with fewer degrees of freedom (we will discuss the role of embodiment in more detail later in this paper).

In conclusion, each discipline that includes communication in its area of interest has explanatory targets that depend on its general nature and goals. Models are designed to address various aspects of communication and language emergence, sometimes using different techniques. AI is the domain in which we see the greatest proliferation of models and in which most novel techniques appear and the domain where research is often motivated by concrete engineering problems and not the theory of language per se. Novel modelling techniques are applied across domains; however, they are already tinted by the specific assumptions adopted within the AI community. For instance, computer science treats symbols just as any discrete elements that can be subject to formal manipulations, and this view is often—implicitly or explicitly—adopted in communication emergence models.

Meanwhile, recent advances in the respective fields that are interested in the theory of language emergence (developmental linguistics, language evolution, and enactivism) have seeped into computational models rather slowly. Consequently, designed models often rely

on a simplified treatment of basic problems, which are the core of more theoretically rich approaches. Moreover, modelling results cannot be easily incorporated into constructed theories. One example is the way the term "compositionality" is often treated in the communication emergence literature, which diverges from the original notion present in the philosophy of language (Rorot and Rączaszek-Leonardi 2023). Here, cognitive science, integrating multiple disciplines from linguistics to psychology to anthropology to computer science, could well serve as a bridge between the domains. Importantly, with its recent "ecological turn" toward action-oriented and situated cognition, cognitive science may integrate processes on biological, developmental, and dialogical time-scales (Rączaszek-Leonardi 2003; Christiansen and Chater 2022).

We align ourselves with the positions of such ecological cognitive science. From this perspective, we point out how the lack of flow of ideas between domains might slow down the progress of computational modelling as a significant aid for the theoretical models of communication and language emergence. In the following, we describe concrete examples of models which best illustrate the various theoretical goals. We will show their advantages and contributions. However, we will also indicate that some of the simplifications they make concern the basic assumptions about the process of communication and are incompatible with theories of language that put greater emphasis on interactions within the environment.

## 3 From Lewis' signalling game to embodied simulations

Despite having different areas of focus, all of the above domains use agent-based models to simulate communication emergence. As a blueprint for the analysis of all kinds of modelling efforts, we adopt the LSG. It provides a minimal interactive communicative setting and delimits a broad class of models, capturing the essential elements of the problem of convention—or signal-based social coordination. It is recognised and sometimes explicitly built upon across various disciplines, from philosophy and anthropology to psychology, biology, economics, and so on. Choosing a common blueprint further facilitates the comparison of the models on a conceptual level in terms of modifications to this basic scheme for various reasons, which can be evaluated as welcome or unwelcome from a theoretical point of view. Showing their basic compatibility with the LSG should also reveal that they share limitations inherent in the blueprint itself if such limitations are present.

The LSG consists of two agents, a sender and a receiver, a set of states of nature (stimuli), a set of signals, and a set of responses. A sender observes a state of nature, sends a signal to the receiver, and the receiver chooses a response. There is a unique correct response given the state of nature unknown to the receiver. If the receiver chooses the correct response, both the sender and the receiver are rewarded. In the course of learning (or evolution), the agents converge to a Nash equilibrium, which could be either a perfect solution when the communication is unambiguous and the receiver always chooses the correct response or a partial pooling equilibrium when different objects are signalled with one signal. The perfect solution occurs when the agents have agreed upon an unambiguous mapping from states of nature to signals and from signals to responses. Lewis called such solutions *signalling systems*.

The original interpretation of the LSG involved rational agents maximising their reward. Skyrms (2010) later reinterpreted it in terms of evolutionary game theory: he demonstrated

that signalling systems may emerge through simple replicator dynamics (which may represent both biological and cultural evolution). He proposed that the information present in the signalling system is sufficient to provide a general account of semantic meaning. According to Skyrms, meaning emerges in the symmetry breaking of converging to one signalling system (out of several possible systems).[6] Each signal can be assigned a descriptive meaning (when an external observer estimates the probability of the sender perceiving a certain stimulus given the signal) and indicative meaning (when an external observer estimates the probability of the receiver performing a certain action given the signal). Such meaning is a property of the overall signalling system and does not presuppose that the sender and receiver are rational agents introducing any kind of meaning of their own.

Several other problems in philosophy and social science can be formalised in terms of LSGs. For example, the labour market can be modelled in terms of employees (senders) who signal their ability to employers (receivers) through educational credentials (signals) (Spence 1973). Similarly, voter ignorance and the spread of misinformation can be modelled in terms of people (senders) maintaining or spreading false beliefs (signals) to signal their group identities or political affiliations (Funkhouser 2017).

Barrett and Skyrms (2017) considered generalised LSGs consisting of multiple subgames organised through a self-assembly. As a simple example, they give alarm calls emitted by animals. At first, a particular alarm call is emitted in the presence of a specific predator. Then, upon hearing an alarm call, an individual might learn to repeat the same call to warn others even without spotting the predator. Barrett and Skyrms discuss two mechanisms involved in the self-assembly process. First is called template transfer, which involves using adaptations that emerged in the context of one game in another game. Template transfer assumes the historicity of the evolution or learning process giving rise to signalling systems—previous adaptations pave the way for the new ones. The second mechanism is modular composition, which is based on recursively nested games, in which a simpler game becomes a part of a more complex one. The two mechanisms are not mutually exclusive, and usually work together. A variety of social, cognitive, and semiotic phenomena can be understood in terms of generalised signalling games, including logical inference (Barrett and Skyrms 2017; LaCroix 2022), information integration in a network of agents (Barrett et al. 2019), functional specialisation, and communication structuring (Barrett et al. 2018).

### 3.1 Beyond Lewis' signalling game

Most of the recent models of emergent communication can be seen as extensions of the LSG concerning particular elements: stimuli, signals, responses, or reward/learning mechanisms (which all necessitate changes in agent architecture). Below, we review various models, from the simplest extensions to complex embodied models, which significantly diverge from the original structure of the LSG. We do not intend to provide a comprehensive review. Our choice of models is dictated by their popularity and their relevance to the problem of the emergence of symbolic communication.

In our overview, we discuss how these models aim to explain different sources of communication complexity connected to a) the internal structure of signals, b) the structure of the environment, c) the structure of actions in the environment and types of control intro-

[6] Here, "symmetry breaking" means that initially, all possible solutions are equiprobable, but once a slight tendency appears to favour one signal over the other the system starts evolving towards a particular solution.

duced by signals, and d) the structure of rewards. The selected models and their properties are presented in Table 1. In the next sections, we discuss them in detail starting from relatively simple formal models and moving towards embodied simulations in realistic environments. Throughout, we refer to the LSG as a blueprint and demonstrate how the presented models deviate from its structure (as reflected by section titles).

### 3.2 Compound signals and noise

The first step in studying interesting phenomena related to natural language evolution is to extend the LSG by adopting compound signals consisting of multiple tokens. A good example is the influential work by Nowak and Krakauer (1999), who studied the dynamics of the evolution of mono- and polysyllabic words and two-word expressions. They investigated whether compound expressions would be evolutionarily preferred to simple words using evolutionary game theory. They constructed a fitness function related to communicative success and identified language structures that optimise this function. The function included factors such as perceptual noise in the signal space (risk of confusing similar syllables) and uneven payoffs for successfully communicating different concepts (for example, signals warning against dangers of different magnitudes). The authors demonstrated that smaller and partially ambiguous lexicons are optimal under certain circumstances. This result might explain why the vocabulary size in natural languages does not grow ad infinitum but stabilises at a certain optimal size.

They further studied compound expressions formed according to rudimentary "grammar", which restricted the space of valid utterances. These compound expressions allowed the agents to exceed the previous vocabulary limits without a negative impact on communicative success. All simulations were kept at an abstract level. The authors demonstrated what sorts of languages are optimal but did not simulate the emergence of actual communication properties. Their work is a valuable demonstration of the role of noise and the complex structure of tasks that lead to uneven rewards in shaping the language structure. These factors align well with our ecological perspective and inspired subsequent work (Kuciński et al. 2021).

### 3.3 Compositional signals via vertical transmission

The interaction between sender and receiver in the standard LSG is horizontal in that, initially, neither the sender nor the receiver knows any meaningful signals and the convention emerges between them. In terms of language evolution, this may represent emerging conventions between peers, but not cultural language transmission from one generation to another. In order to target this phenomenon, Kirby (2001) developed an iterated learning model, which inspired a large body of research over the years (Kirby et al. 2014).

In the iterated learning model, a student (receiver) observes a set of stimulus-signal mappings generated by a teacher (sender) from the previous generation. Signals are compound multisyllabic utterances, and stimuli are objects described by the composition of stable properties. The produced stimulus-signal pairs become a learning sample for the next sender, which produces a learning sample for yet another sender, and so on. When the student is unable to learn all stimulus-signal pairs from the previous generation perfectly, vertical transfer introduces a bottleneck shaping the structure of language. Iterated learning

**Table 1** Models of emergent communication discussed in the paper, their origin in various traditions, and their characteristics in terms of the basic template of the LSG. The models were selected to display various interaction scenarios considered as communicative situations in the model. We ordered them according to their abstractness from idealised formal models to embodied simulations/mobile robots

| Model | Discipline | Environment (stimuli) | Signals | Available actions | Rewards and learning mechanism | Results/ explanatory target |
|---|---|---|---|---|---|---|
| Lewis (1969) | Philosophy | Discrete tokens | Discrete tokens | Discrete tokens | A specific response rewarded | Stabilisation of social conventions |
| Nowak and Krakauer (1999) | Linguistics | Discrete tokens | Sets of tokens (compound expressions), risk of confusion | Discrete tokens | Different rewards are given depending on the response | Evolutionary origins of phonological and grammatical structures |
| Smith et al. (2003) | Linguistics | Binary vectors: combinations of properties | Sets of tokens (compound expressions) | – | Learning mapping after the previous generation in a supervised manner | Origins of systematic structure |
| Batali (1998) | Linguistics | Binary vectors: simple predicates | Sequences of tokens (compound expressions) | Binary vectors: reconstructed predicates | The sender tries to predict the receiver's responses internally (obverter). The receiver learns from the sender in a supervised manner. Roles change periodically. | Emergence of systematic structure in sequences of variable length |
| Jaques et al. (2019) | Artificial intelligence | 2D grid world with simple objects encoded as binary vectors | Discrete tokens | Movement in the environment | The number of collected food items is rewarded, group reward | Emergence of communication under intrinsic social influence motivation |
| Steels and Belpaeme (2005) | Cognitive psychology | Sets of numerical vectors representing colours (available to the sender and receiver), marked stimulus (available to the sender) | Discrete tokens from an infinite set | Choosing the correct stimulus | Specific response rewarded, supervised learning of new word forms | Emerging communication stabilising category formation |

**Table 1** (continued)

| Model | Discipline | Environment (stimuli) | Signals | Available actions | Rewards and learning mechanism | Results/ explanatory target |
|---|---|---|---|---|---|---|
| Lazaridou et al. (2017) | Artificial intelligence | Sets of numerical vectors: raw RGB images | Sets of tokens (compound expressions) | Choosing the correct stimulus | Specific response is rewarded | The signalling system is grounded in raw visual input |
| Cangelosi (1999, 2001) | Linguistics & philosophy | 2D grid world with objects encoded as binary vectors (available to sender and receiver; the receiver gets perceptual noise) | Sets of tokens (compound expressions) | Discrete movement in a grid-like environment and discrete response tokens | Locating food and avoiding poison are rewarded in evolution, the speaker is not rewarded | Symbolic mode of reference - transfer of symbol-symbol relations to new vocabularies |
| Schulz et al. (2012) | Robotics | 2D realistic environment, own position in the environment (available to the sender and the receiver) | Sets of tokens (compound expressions) | Continuous movement in the environment (available to sender and receiver) | Meeting at the appointed location is rewarded | Constructing abstract location concepts grounded in spatial experience |
| Loula et al. (2010) | Linguistics & philosophy | 2D grid world, 0–1 presence of nearby objects of various classes (available to all agents) | Discrete tokens from a large set | Choice of high-level, predefined behaviour: wandering, fleeing, etc. (available to all agents) | Hebbian learning of associations (classical conditioning) | Shift between different forms of signification within an ecological niche |
| Mordatch and Abbeel (2018) | Artificial intelligence | 2D environment, (shared by the sender and the receiver), marked target object and action (available to sender) | Sets of tokens (compound expressions) | Continuous movement in the environment | Explicit error feedback | Systematic signal structure is grounded in action structure |

**Table 1** (continued)

| Model | Discipline | Environment (stimuli) | Signals | Available actions | Rewards and learning mechanism | Results/ explanatory target |
|---|---|---|---|---|---|---|
| Floreano et al. (2007) | Robotics | 2D environment (shared by all agents) | Light emitted by physical robots | Continuous movement in the environment (available to all agents) | Locating food and avoiding poison are rewarded in evolution, the speaker is not rewarded | Communication strategies evolve under different kin structures |
| Grouchy et al. (2016) | Robotics & linguistics | 2D environment (shared by all agents) | Continuous scalar | Continuous movement in the environment (available to all agents) | No explicit rewards, two agents producing offspring upon contact | The emergence of a complex mode of signification with signal-signal relations |

model was mathematically analysed using evolutionary game theory (Brighton 2002) and implemented in computational simulations. Iterated learning intends to explain the systematic structure of natural languages.

Smith et al. (2003) constructed a simulation in which they represented agents as simple neural networks consisting of two layers of binary neurons. The first layer represented binary object properties (each object either had or did not have a particular property). Meanwhile, the second layer represented the presence of specific components in the utterance (each utterance was an unordered set of components). The connections between neurons were reinforced based on the object-utterance pairs present in the learning input. Then, a set of utterances for new objects was produced.

A transmission bottleneck was implemented as follows: agents from the new generation observed only a subset of utterances produced by agents of the previous generation. They demonstrated that under these circumstances compositional languages—in which individual syllables in an utterance are systematically mapped to object properties—emerge. Importantly, Kirby et al. (2008) replicated the same schema in an experiment involving human participants and obtained similar results.

These studies consistently show that a transmission bottleneck introduces evolutionary pressure for selecting languages that are relatively easy to learn, which leads to the structurisation of meanings. Structured meanings are fundamentally different from the compound expressions with protogrammar analysed by Nowak and Krakauer (1999). In that work, grammar defined only the structure of an utterance, demanding that the word of a particular class appears at a particular position in a sentence, which was in principle independent of the word's meaning. The structure of the compound expressions in Smith et al. (2003) was not restricted, but the way agents referred to the properties of objects was systematic: different parts of the expression referred to different properties of objects. This can be called compositionality in terms of meanings but without specific grammar (Rorot and Rączaszek-Leonardi 2023).

Models based on vertical language transmission can be seen as orthogonal to the horizontal LSG. They do not include a coordination problem, but rather a memorisation problem:

agents learn the signalling protocol by observing examples rather than through interaction. Vertical and horizontal approaches can be seen as complementary, and some models explicitly combine vertical iterated learning with the horizontal LSG. Such a combination of vertical and horizontal transfer was implemented by periodically resetting the weights of the neural networks of some agents (clearing their memory; Li and Bowling (2019); Cogswell et al. (2020)) or by having the agents passively observe the communication protocols of the previous generation (Ren et al. 2020).

### 3.4 Social reasoning facilitating communication

A different line of work looks for pressures for structured communication in pragmatic social reasoning, which is involved in language generation and comprehension. In formal linguistics, this kind of reasoning is usually approached using the Bayesian Rational Speech Act framework (Frank and Goodman 2012). The central idea is to introduce a pragmatic sender, which takes into account the receiver's perspective and chooses a signal that has the highest probability of being correctly understood. Then, a pragmatic receiver may consider the pragmatic sender's perspective, and so on.

We recognise that a similar idea was introduced in the communication emergence literature by Batali (1998), who attempted to explain the origin of the structure of natural languages. His simulations preserved the basic sender-receiver structure of the LSG, but the sender used its own language comprehension module to predict the receiver's responses. Based on these predictions, it generated a signal that would be easy to interpret—an approach dubbed "obverter" (Oliphant and Batali 1997). The population of agents learnt in a self-supervised manner: a pair of randomly selected agents interacted in each round. The sender acted as a teacher for the receiver, and, importantly, the roles were periodically swapped. The language that emerged had compositional properties similar to the language emerging in the experiments by Smith et al. (2003), as some parts of the signal sequence had systematic meanings. Furthermore, the agents were able to generalise and communicate novel combinations of meanings not present in the training set.

The obverter mechanism uses the same signal-meaning mapping for language generation and comprehension. This guarantees that agents are internally consistent and use the language they learnt in the receiver role when they assume the speaker role. It can be interpreted as a rudimentary form of social reasoning: the sender produces the signal, which has the highest probability of mapping to the desired object if the sender itself were to interpret this signal. Imagine that the receiver is the source of the language, and the sender attempts to learn the language of the receiver (the obverter mechanism makes the sender explicitly adjust to the receiver). Within a single round of the game the sender is able to learn only a single association at best, which makes language transfer imperfect. As the roles are assigned at random, in the next round, the sender may become a new receiver and a source of language for a new sender, further propagating its learning biases. Exchanging sender-receiver roles effectively creates a transmission chain with a learning bottleneck similar to the iterated learning setting (Smith et al. 2003). Such a transmission chain constitutes a pressure for a systematic communication protocol. The obverter mechanism was implemented in modern neural networks by Choi et al. (2018) and Bogin et al. (2019), who also observed an increase in compositionality.

More complex mechanisms of social reasoning have also been explored. A good example is the work of Jaques et al. (2019), who introduced the concept of social influence as an auxiliary reward for agents communicating to solve a sequential social dilemma. The authors presented a measure of social influence which incorporated counterfactual reasoning: an agent considered the alternative signals it could have sent and the plausible actions the other agent could have taken in this alternative world. Based on that, the first agent assessed whether actions of the other agent were contingent on its signalling or would have been taken regardless of the signal value.

In one variant of their simulations, agents' used their own assessment of whether their signals were influential to estimate their social influence reward. In order to achieve this, one agent predicted the actions of the other agent using a specialised internal model trained in a supervised manner. Unlike in Batali (1998), the model of the other agent was separated from the main model that controlled the actions of the agent. The authors did not focus on the signal structure, but showed that supporting communication with counterfactual reasoning leads to faster convergence, higher cumulative rewards, and higher mutual information between signals and actions. Even if agents did not have an explicit communication channel, influence rewards incentivised them to develop an ostensive communication protocol by pointing out important locations by staying still at these locations. This pointing behaviour emerged because, from the beginning, agents viewed their actions as means to not only achieve their immediate goals but also to influence others. The dual meaning of the action—performative and communicative—followed the dual structure of the reward function including both task performance and social influence.

The results of these simulations suggest that some aspects of social reasoning are necessary to introduce pressures for communication relevance and the ease of interpretability (in line with Grice's maxims of conversation (Grice 1975)), which drive language evolution. Consequently, agents might require certain cognitive preadaptations for social cooperation or scaffolding in the form of existing social relations in order to establish a nontrivial communication protocol.

### 3.5 Bidirectional influences between perception and communication

Harnad (1990) posed the famous symbol grounding problem, which asks how symbols (which he understood as elements of the symbol system that are subject to formal syntactic manipulations) can refer to anything in the outside world. He suggested that when natural languages are being developed and learned, symbols are initially grounded in basic perceptual processes, and then more abstract learned or innate categories are formed around perceptual features.

Among works in communication emergence that explicitly address the symbol grounding problem, the work of Steels and Belpaeme (2005) stands out as particularly comprehensive. The authors used a variant of a language game, which they called a guessing game. Similar to the standard LSG, it involved a sender and a receiver communicating the properties of the presented stimuli: vectors of real numbers representing colours. In contrast to the standard LSG, in which there is no context, here the sender was presented a single "target" stimulus in the context of three other "distractor" stimuli. The stimuli were drawn at random. The only restriction was that the target had to be sufficiently distinct from distractors–this was determined by the size of just-noticeable difference between colours as estimated

in human-subject studies. The receiver was presented with both the sender signal and the same set of stimuli, and its task was to correctly choose the target stimulus. Each agent used a lexicon module—a mapping between discrete categories and words—and an adaptive categorisation module responsible for forming categories in a continuous perceptual space. The lexicon was updated whenever a previously unknown word was encountered. Successful communication required the coordination of both vocabulary and perceptual categories. In the simulation, a population of agents interacted randomly with each other freely exchanging sender and receiver roles. No colour categorisation was designed as "correct".

The authors demonstrated how a conventionalised categorisation of the colour stimuli may emerge through linguistic interaction. The vocabulary was also shared within the population and-through the categorisation module-grounded in the perceptual properties of the stimuli. Similar models were used to study how aspects such as innate perceptual biases (Baronchelli et al. 2015; Ohmer et al. 2022) or the topology of a social network (Centola and Baronchelli 2015; Zubek et al. 2017) affect the formation of categories. Further work by Steels (2015) brought guessing game schema to the context of developmental robotics, where simulated agents were replaced with physically embodied robots. Spranger (2016) extended the model and used it to demonstrate the emergence of spatial language. These models are unique because they demonstrate that both signal-category associations and the perceptual categories themselves can be subject to conventions formed bottom-up through interaction.

### 3.6 Signals grounded in raw sensory inputs

The advent of convolutional neural networks made it feasible to introduce more realistic visual stimuli into communication emergence models, substituting discrete tokens from the basic LSG with images. Some of the first works exploring this path were conducted by Lazaridou et al. (2017) and Lazaridou et al. (2018). The departure from pre-processed symbolic tokens input (e.g. encoding an object as a binary vector) in favour of raw visual input (i.e. arrays of RGB pixels) requires the sender to deal with entangled input and decompose it into relevant factors of variation (e.g. shape and colour) on its own. Lazaridou et al. (2017) demonstrated that a model trained using reinforcement learning learned to successfully play a referential game (although, it was not fully trained end-to-end as it used a pretrained module to extract image features). Moreover, Lazaridou et al. (2018) observed that when compound signals were used, emergent communication protocols were, on average, more systematic for symbolic object representations than for raw pixel inputs. Communication protocols emerging with visual input did not always capture interpretable, concept-level properties of visual scenes presented to the agents. Instead, agents learned to successfully distinguish images while overfitting to the inscrutable low-level relational properties of images.

Bouchacourt and Baroni (2018) demonstrated that agents can solve guessing games with high accuracy even when the input data contains no conceptual content at all-for example, when images comprise random noise sampled from a Gaussian distribution. Researchers discovered ways to foster more systematic communication protocols with visual input, for instance, by increasing the number of available targets (Mu and Goodman 2021). However, the overall conclusion is that realistic visual input alone does not contribute to the structure of language. Systematic structures must be present in the way agents perceive the environment.

### 3.7 Multi-task learning: actions beyond signalling

Embedding the LSG in a realistic context requires that the agents are able to perform actions other than emitting signals. Then, the structure of the emerging communication protocol could be influenced by the demands of simultaneous signalling and acting in the environment. Among works exploring this matter, we have chosen the model presented by Cangelosi (1999, 2001) as particularly interesting, since it grounds computational simulations in a deeper semiotic theory produced by Deacon (1997) (which is his own reinterpretation of Peircean semiotics). Deacon views communication emergence as a multi-stage process, during which one type of sign is transformed into another. He distinguishes between iconic signs grounded through perceived similarity, indexical signs grounded in spatio-temporal contingencies, and symbolic signs, which are grounded primarily in relations with other signs. According to his proposal, indices can be transformed into symbols through weakening sign-referent associations (think: word and concrete objects in the world) while strengthening sign-sign associations (think: syntactic rules). Symbols can thus be viewed as denoting more abstract relations that can be transferred between contexts.

Cangelosi attempted to simulate this process in a population of agents foraging for mushrooms of different types. Agents were rewarded for correctly identifying a mushroom type based on unreliable perceptual input and a signal produced by their peers. The identification meant correctly classifying a mushroom as either edible or inedible, and, in the case of edible mushrooms, classifying its subtype. Each agent was controlled by a separate neural network. Networks were evolved through a simple genetic algorithm (population learning) and fine-tuned using backpropagation (individual learning). The agent network had separate output layers responsible for identifying mushrooms and producing signals, and a hidden layer was shared between these two tasks.

Communication had a structure similar to the basic signalling game, in which an agent foraging for mushrooms was the receiver, and its parent from the previous genetic algorithm generation was the sender. The sender was presented with a true vector of perceptual properties of a given mushroom and produced a two-component signal. The receiver was presented with a noisy version of the vector of mushroom properties and the sender's signal. It had a double task of correctly identifying the mushroom (rewarded in the evolutionary timescale) and recreating the sender's signal (trained with backpropagation in a supervised manner). This elaborate setup led to the emergence of a systematic communication protocol with disentangled mushroom properties.

To check the generalisation capabilities of these systems, Cangelosi applied a test based on Deacon's interpretation of the classic experiments of teaching language to chimpanzees (Savage-Rumbaugh and Rumbaugh 1978; Savage-Rumbaugh et al. 1980). The rationale for this is as follows: let us assume that we know some basic vocabulary in one language. Then, some new foreign vocabulary is introduced in a grounded manner, albeit for a limited number of examples. Next, the learner is presented with combinations of words in the foreign language that allow them to recognise a familiar grammatical rule. Afterwards, the learner should be able to correctly use the foreign words in combinations, even if they were not introduced explicitly for all examples.

This schema was replicated 1–1 with the neural networks. The trained network represented the properties of mushrooms A and B with independent symbols. Cangelosi retrained the network using backpropagation to substitute existing symbols for property A for two-

thirds of mushrooms with new symbols. Then, he introduced new symbols for property B and taught the network how symbols for properties A and B are composed together in two-component signals. Finally, he introduced new symbols for property A for the remaining third of the mushrooms. At this point, the network was able to correctly associate all symbols A and B even though this association was explicitly given only for two-thirds of mushrooms.

This kind of generalisation goes beyond the generalisation of responses to new stimuli; this is a systematic transfer of relations formed between one group of symbols to a new group (in accordance with Deacon's semiotic framework). Using the terms of Barrett and Skyrms (2017), we consider this an example of template transfer, by which the stimuli remain the same but the response repertoire changes. Cangelosi et al. (2002) interpreted this generalisation as a process of "symbolic theft", by which preexisting grounded symbols and the relations between them are used to ground new ones. We suspect that the agents' ability to transfer the structure of relations to new symbols is a consequence of the simultaneous involvement of both sender and receiver in classification and signalling tasks. This involvement forced them to disentangle relevant factors of variability in the environment in their internal representations. However, this was not tested directly in the original work.[7]

### 3.8 Modelling an ecological niche for communication

Some of the models inspired by artificial life research have focused more on the role of the ecology and how it may shape a niche in which communication evolves than on the architecture of or interactions between individual agents. One study of this kind was conducted by Loula et al. (2010). They were inspired by the ecology of vervet monkeys-animals with a particularly sophisticated set of alarm calls. In the simulation, a group of herbivore creatures explored a 2D environment looking for edible plants while avoiding predators. Contrary to the basic LSG, signalling behaviours existed within the natural ecology of the simulated environment, going beyond static sender-receiver interactions. Simulated creatures could see nearby objects, hear vocalisations, and select predefined actions based on their perception and internal drives (hunger, fear, boredom, etc.). Actions were high-level behaviours, such as scanning the environment, wandering, fleeing, hiding, climbing a tree, and vocalising. The communication was not limited to sender-receiver pairs: any creature within the vicinity was able to hear a vocalisation and react to it (possibly with another vocalisation).

No particular reaction was distinguished as "correct". Systematic associations between vocalisations (signals) and predator types (meanings) were spontaneously self-organised through Hebbian learning of associations between stimuli in this ecology. The authors demonstrated how the emergence of signals transforms ecology. At first, creatures reacted to signals by scanning their environment for potential predators: all signals were interpreted as general alarm calls signalling danger. Later, when signals were well learnt, the creatures took actions adequate for the predator type (fleeing, hiding, or climbing a tree) immediately after hearing the signal without ever seeing the predator. Moreover, the creatures repeated the signal they just heard to warn the others. This is consistent with the example of signalling game self-assembly given by Barrett and Skyrms (2017).

[7] In Cangelosi's simulations, agents also moved in a 2D environment and had to approach a mushroom before picking it up. However, since they did not have any memory, their movement trajectory did not influence their classifications or contribute to the structure of their communication.

Loula et al. (2010) demonstrated a shift between different forms of signification, by which the same signals gradually changed meaning and started referring to entities not directly present. Such a shift is possible by grounding communication in a sufficiently rich ecology. Complexification of an ecological niche might be a necessary step on the road towards symbolic communication from simpler signals.

### 3.9 Embodied and situated communication

In the vanilla LSG, all actions are discrete: the model abstracts away from the dynamics of action in a continuous environment. If we were to take embodiment more seriously, in line with the proposition of ecological approaches to cognition (Rączaszek-Leonardi 2024), we would have to consider the spatio-temporal characteristics of the processes of acting and signalling as viable sources of communication structure. We have selected two examples of works pursuing this direction.

The first is the work by Mordatch and Abbeel (2018), which is well-known in the AI community. In their simulation, a 2D continuous space was populated with agents and coloured landmarks. Agents could move in the space and emit discrete signals, and their policies were represented by a recurrent neural network (all agents shared the same policy network). Each agent was given private information about the action the other agent must undertake, so agents had to instruct one another. The agents were trained with backpropagation to communicate to the other agent the correct action (e.g. GOTO) and object that this action must target (e.g. BLUE).[8] The communication protocol that emerged in the simulations had a compositional verb-noun structure. Compositionality was encouraged through a soft limit on vocabulary size that made agents reuse already known words (introducing a new word was penalised).

The ability of an agent to move freely contrasts with dominant settings, according to which the sender passively observes a stimulus (independently of any actions it can undertake) and the receiver waits for a message to undertake an action. The fact that the sender could move enabled nonverbal communication as the receiver used, for example, the gaze direction or physical movement of the sender to inform its actions. Moreover, according to the authors, the physical properties of the environment and agent communication (e.g. actions and utterances that take time to accomplish) shaped the syntactic structure of the utterances. In the communication protocol that emerged, the GOTO action type signal was uttered before the landmark signal, allowing the receiver to "prepare" for further actions by moving to an advantageous central position among the landmarks before starting to move towards the specific one.

The second example of a work exploring a rich and structured environment as a potential source of structure appearing in language comes from the robotics tradition. Floreano et al. (2007) employed a swarm of mobile robots that navigated their way through a 2D space to discover a food source and avoid a poison source. The robots were controlled by neural networks evolved through an evolutionary algorithm, in which fitness was determined by their

[8] The authors remarked that agents could, in principle, be trained using reinforcement learning, but for practical reasons, they decided against it. Instead, they built a fully differentiable system consisting of two agents and their environment, and they trained it end-to-end using backpropagation. Gumbel-Softmax was used to make the discrete communication channel differentiable. The training was supervised in the sense that the training signal contained information about correct actions, but it did not contain information about correct signals - goal-signal associations emerged spontaneously.

food intake. They could emit light signals, which helped other robots locate food, but this created competition for resources: spatial limitation of the environments allowed only a limited number of robots to feed at the same time. Contrary to the standard LSG, there were no sender-receiver roles, no turn-taking, and no explicit reward for successful communication. The results indicate that cooperative communication can emerge among kin groups even if signalling is costly for individuals. More varied behaviours, such as the emergence of deceptive communication in groups of nonrelated robots, were also observed. Importantly, the values and costs of communication in these experiments were not arbitrarily introduced but emerged from the properties of the environment.

Based on these two examples, we conclude that embodied simulations add many degrees of freedom to the model, making it more difficult to analyse while also allowing for new, often unexpected, pressures that enable communication structure to emerge. Such pressures might be essential for moving towards symbolic communication.

### 3.10 Combining concepts across games

Some works in robotics examine communication emergence in even more structured environments. As an example, we present the work of Schulz et al. (2012), which is both technically sophisticated and theoretically ambitious. They experimented with mobile robots, which navigated a natural environment (office space) and learned to refer to specific locations within that environment. The robots could move freely in their environment and construct private maps of the space based on visual stimuli and self-motion odometry (using the RatSLAM algorithm (Milford and Wyeth 2010)). The map for each agent was slightly different, as it reflected each agent's individual experience.

When two robots met, they were able to hear unique signals that they emitted, and they engaged in sender-receiver guessing games inspired by Steels and Belpaeme (2005). The structure of these games was explicitly programmed by the authors and was independent of the agents' motor activities. The only aspect that was modified through the games was the association map between signals and spatial locations in the private map. The new element was that the agents played a sequence of progressively more structured games and that the concepts acquired through these games were interrelated.

They started by playing a where-are-we game in which they named the location of their meeting. After a few games, the agents had conventionalised the lexicons of location names grounded in their private spatial maps. Then, the researchers made the robots play two new games, how-far and what-direction, using grounded location names. In the how-far game, two locations were named and the robots had to negotiate words to express the distance between them. In the what-direction game, three locations were named and the robots coordinated terms to express an angle between the two locations from the point of view of the third. After they coordinated their lexicons, they were able to use both location names and relational vocabulary to successfully play go-to games when they arranged to meet at a specific location. Then, in the where-is-there game, the agents used the relational vocabulary to name the locations they had not yet visited. The labels established during this last game remained abstract until the agents were able to discover previously unknown locations through exploration.

The achievement of this model is a particularly clear presentation of two different methods of grounding vocabulary. The first was direct grounding in the experiences of spatial

locations. The second was indirect grounding in the relations between already established concepts. Schulz et al. (2012) demonstrated how this more advanced type of grounding can be achieved when simpler games are combined using a modular composition schema (Barrett and Skyrms 2017). Combining games required a progressive complexification of agent interactions (here, explicitly encoded in the structure of guessing games) and a conceptual space allowing meaningful operations on concepts (here, spatial maps of the environment). Such double grounding is a compelling idea in cognitive science and may be indispensable for abstract concepts to emerge (Rączaszek-Leonardi and Zubek 2023).

### 3.11 Situated coordinative dialogue

The LSG is structured to portray a descriptive function of communication: the sender describes the state of affairs unknown to the receiver. The communication is unidirectional and follows knowledge asymmetry. Only a few models depart from this assumption. Among them, we recognise the work of Grouchy et al. (2016), which stands out due to the original modelling techniques used. The authors modelled a scenario involving no external states to communicate; the scenario only contained agents that coordinated their movements in a continuous 2D space to meet each other.

They simulated colonies of mobile robots with controllers evolved through genetic programming (evolved controllers were later transferred to physical robots: e-pucks). The controllers took the form of ordinary differential equations that defined the relationship between robot sensors and actuators. The robots could not see each other but could hear signals emitted by nearby peers. The robots communicated by emitting sounds (the communication channel was continuous and one-dimensional). Whenever two robots met in space, they crossed their genetic material and produced offspring. There was no explicit reward or fitness function, but the reproduction mechanism favoured robots that were able to find each other more efficiently, which, in turn, applied evolutionary pressure on emitting and interpreting sounds, thereby developing an efficient communication system.

Most populations of robots evolved communication systems in which the emitted signals corresponded to the robot's position in one of the two dimensions. Some populations used this communication strategy for a while but later on evolved more complex strategies, by which the signals emitted by two robots became co-dependent: a signal emitted by one robot was modulated by the signal emitted by the other robot. The dialogical communication established between the two robots became a dynamic control structure, which facilitated their meeting. The two modes of communication depended on each other since the dialogical communication strategy emerged from the simpler one. Conceptually, this resembles the modular composition of signalling games proposed by Barrett and Skyrms (2017), although here, the signalling structure is fully emergent, without explicit roles, rewards, and so on.

Grouchy et al.'s model remains original, as it challenges many assumptions common in communication emergence models: a) it uses continuous signals instead of discrete signs, b) it does not reward agents explicitly, and c) it explores a form of communication that is less about representing states of the world and more about the dynamic coordination of behaviour. Concerning b) and c), the model by Floreano et al. (2007) is similar.

## 4 Emergence, function of communication, and the mapping metaphor

The structure of existing models reveals commendable efforts to investigate various facets of the communication emergence problem. These models have demonstrated stable communication protocol formation under diverse circumstances (Spike et al. 2017). The protocols are efficient, as they enable agents to carry out tasks effectively, and some simulations highlight the benefits of explicit signalling compared to a situation without a designated communication channel (e.g. Jaques et al. 2019; Mordatch and Abbeel 2018). Emergent protocols often allow for generalisation to new contexts (e.g. Cangelosi 2001; Schulz et al. 2012) and display a certain degree of systematicity (or compositionality) (e.g. Mordatch and Abbeel 2018; Chaabouni et al. 2020).

The iterated learning model provides a convincing explanation for the basic compositionality of natural languages as a solution to the transmission bottleneck and language learnability problem (Kirby et al. 2014, 2015). Recent advancements fuelled by the implementation of deep neural networks (Lazaridou and Baroni 2020; Chaabouni et al. 2022) enhanced individual agents' cognitive capabilities, which contributed to more biologically authentic simulations. Agents are now able to handle more realistic inputs such as images (Lazaridou et al. 2018), navigate more intricate environments, including social environments (Batali 1998; Jaques et al. 2019), and align their signals with basic English vocabulary (Lazaridou et al. 2020).

Despite these advancements, some questions about the emergence and evolution of symbolic communication remain open. These include the emergence of discrete signs from ordinary actions in interactions, the trajectory necessary for such signs to evolve into symbols embedded within a network of symbol-symbol relations, the contextual plasticity of symbols, and, in connection with this, non-trivial compositionality (aspects of compositionality beyond the logical intersection of two sets (Steinert-Threlkeld 2020)). The adoption of specialised neural network architectures, such as convolutional neural networks (LeCun et al. 1989), long short-term memory networks (Hochreiter and Schmidhuber 1997) or transformers (Vaswani et al. 2017) have allowed agents to process more realistic inputs or longer sequences of tokens. However, this alone does not address the fundamental questions above.

It is telling that, while a great many works use the term "language emergence" or "communication emergence", only a few have attempted to define this term more precisely, apply a specific theory of emergence, and discuss which aspects of the simulation are emergent and which are straightforward consequences of the assumed model structure (Di Paolo 1997; Hutchins and Johnson 2009; Loula et al. 2010). Taking this into account, we suggest that progress is hindered not by the lack of sophisticated computational architectures but by the lack of reflection upon some basic assumptions behind language emergence modelling.

We identify several gross categories of simplifications, which-while useful in addressing concrete engineering problems or concrete isolated theoretical questions-might obscure the essential properties of symbolic communication that need to be addressed in a successful explanatory model:

1. Lack of emergence. In existing models, communication rarely truly emerges, as symbols are predefined and given as discrete numbers or vectors. They usually form a separate modality distinct from the perceptual input, which is already designed for communication by the model structure. The problem of the conventionalisation of discrete signals

and symbols emerging from interactions is excluded. The existence of discrete symbols is presupposed rather than explained. Agents' actions also tend to be discrete and devoid of dynamics, preventing the emergence of new forms of organisation. This assumption is somewhat relaxed in only a couple of models (pointing and "gaze" direction in Mordatch and Abbeel (2018), movement and use of light properties for communication in Floreano et al. (2007), and continuous signals and movement in Grouchy et al. (2016)).

2. Limiting the function of communication. In most of the existing models, communication has a single function which is explicitly rewarded. The focus on the descriptive function of communication is evident in settings where there is knowledge asymmetry between agents, and one agent is rewarded for passing certain information about the world to the other. The imperative or coordinative function of communication concerning the regulation of the ongoing interaction (for example, by eliciting certain behaviours of the interaction partner) is subordinate to the descriptive function (as in Schulz et al. (2012), where robots first learn descriptive communication and then use it imperatively). A notable exception is the model of Grouchy et al. (2016), which demonstrates how coordination can be independent of the description of the external environment.
3. Mapping metaphor. The meaning of language is reduced to mapping discrete signals to discrete referents. The meaning-mapping assumption is visible in the basic LSG, in which the reward is maximised when the emerging signals can be uniquely mapped to the states of the world. Compositionality is understood in terms of more complex structured mappings (Korbak et al. 2020). In the simplified environments adopted by most simulations, actions performed by agents and action history do not significantly contribute to the meaning of communication. An effort to break free from the mapping metaphor is visible in Loula et al. (2010), where emerging meaning is understood as a transformation of the ecological niche rather than simple mapping. Moreover, in Grouchy et al. (2016), signals allow functional coordination without mapping to specific referents.All these simplifications are in line with the basic structure of the LSG as seen through the lenses of Shannon's information transmission (Skyrms 2010). They can be viewed as interdependent; for instance, a mapping metaphor assumes the existence of discrete signals and referents, thereby strengthening the notion that they are preexisting rather than emerging in the process of communication as well. Similarly, if we consider the fundamental role of communication to be descriptive, building a word-meaning mapping emerges as a logical solution. Although some existing models depart from these simplifications (usually not all three of them at once) the simplifications hold surprisingly strong in general. We assume that this is due to a widespread approach to symbols, which treats them as something "given", existing independently of meaning, similar to unbound variables in a computer program ready to be associated with some values. Symbols are thus perceived as discrete abstract entities with formal properties that can be manipulated.

Years of dominance of the information processing and computer metaphor in the cognitive sciences have resulted in thinking about symbols as necessary assumptions before constructing explanations. This makes it easy to forget that their existence also needs to be explained. Consequently, models focus on how we can manipulate symbols, combine them, and decipher their meaning or "ground" them. However, the problem of the emergence of complex communication essentially includes the problem of symbol emergence: how do

they come into being in the first place? What kinds of meaningful relations do they require to be described as symbols (Rączaszek-Leonardi et al. 2018)?

Using abstract symbols with arbitrary meanings as a starting point is also limiting because it puts the burden of structuring communication on the conventions created from scratch by the agents and realised by their internal architectures, while the role of interactions with a richly structured environment is neglected or limited. In the real world, forces from multiple sources shape and structure communication. For instance, the physical properties of signals also partly determine their meanings (like the light direction in Floreano et al. (2007)), and situated agents can only communicate effectively within a certain distance (as in Loula et al. (2010)). Neglecting the ways symbols might be created in interactions within a particular ecology obscures these sources of complexity, leaving us only with the simplified view of communication meaning as mapping from symbols to features (Rączaszek-Leonardi and Deacon 2018).

## 5 Sources of communication complexity

Following the above review and the summary of theoretical considerations, we have identified some of the most important sources of communication complexity explored by different models. We discuss which of them could be explored in more depth in order to relax three simplifying assumptions identified in the previous section.

### 5.1 Perceptual grounding

Grounding language in perception sets most emergent communication models apart from distributional semantic approaches, by which language meaning is derived solely from word co-occurrences. In perceptually grounded models, the agent's perception plays a crucial role in shaping language structure. For instance, in Steels and Belpaeme (2005), vocabulary structure was influenced by agents' perceptual capabilities defined in terms of the size of just-noticeable difference between colours. Smaller size of the just-noticeable difference led to a larger number of perceptual categories and larger vocabulary. To explore the effects of perceptual grounding, some models employ realistic stimuli (e.g. images represented as raw RGB pixel arrays, as demonstrated in Lazaridou et al. (2018)), and incorporate realistic perceptual systems, including multimodality, limitations of an agent's senses, and perceptual noise, among others Evtimova et al. (2018); Hoang et al. (2024).

Making an agent's perception more realistic increases the complexity of the model, but it does not guarantee that a more structured and systematic communication protocol will emerge. In fact, as shown in Lazaridou et al. (2018), communication protocols trained with image input were less structured than those with symbolic tokens input. As long as perceived stimuli-however complex-are objects that are mapped to signals, perceptually grounded models do not solve our issues connected with the mapping metaphor.

### 5.2 Sensorimotor grounding

Another level is grounding language not only in passive perception, but in experiences gained from active exploration of the environment (active perception, Gibson (1966)). Such

grounding requires an embodied agent capable of nontrivial actions that affect states of the world that later affect the perception of the agent (closing the sensorimotor loop). Simple extensions of the signalling game model or machine learning models inspired by image captioning or question answering tasks preclude that. Sensorimotor grounding is possible in embodied simulations, such as Mordatch and Abbeel (2018), where agents move in the environment and perform sequences of actions, which in turn opens up additional possibilities for language complexification. Research in robotics has shown that agents with closed sensorimotor loops and relatively simple perceptual input are able to evolve interesting communication systems (Floreano et al. 2007). With sensorimotor grounding, agents can use communication to directly coordinate their actions. More complex sensorimotor grounding can potentially be a way to move away from the meaning-mapping metaphor and from the descriptive function of communication, as suggested by experiments with infants and embodied robots (Wrede et al. 2009; Rohlfing et al. 2016; Lyon et al. 2016).

### 5.3 Properties of sign vehicles

Communication complexity can also be induced by the physical properties of sign vehicles (signals) used in communication. As we observed before, many models utilise discrete and amodal signs and do not explore physicality of signs as a dimension of communication complexity at all (they treat symbols as given, not emergent). Some models follow the typical sender-receiver schema but introduce noise into the signal space (Nowak and Krakauer 1999; Kuciński et al. 2021) or allow for discrete signals to emerge from a continuous signal space (Geffen Lan et al. 2020; Lieck and Rohrmeier 2021)-these could be treated as the first steps in this direction.

In embodied models where signals have physical vehicles, such as the model of Floreano et al. (2007), which uses light signals, the properties of vehicles and the properties of the perceptual systems of agents interact. This interaction imposes new constraints on the agents' interaction scenarios-for example, communication may be possible only in the vicinity of another agent. In addition, physical sign vehicles allow us to explore aspects of iconicity—the similarity between a sign vehicle and its referent (be it an object, action, event, etc.)—which cannot be studied using amodal signs (Hutchins and Johnson 2009). The complexification of sign vehicles is necessary to eliminate the assumption concerning the pre-existence of discrete symbols and allow them to emerge (Taniguchi et al. 2016).

### 5.4 Complexity of the environment

It is generally recognised that a sufficiently complex environment is necessary for complex communication to emerge. Within the basic LSG schema, the degrees of freedom of signals are limited by the degrees of freedom of stimuli. Simulations that deal with natural images as stimuli explore this dimension of complexity. However, as demonstrated by Bouchacourt and Baroni (2018), using complex stimuli does not automatically lead to meaningful conceptual representations in agents.

With the recent success of large language models, some researchers speculate that simply increasing the size and variability of training data could bring the internal representations of models closer to interpretable concepts. Recent works demonstrate that meaningful spatio-temporal and truth-false relations can be recovered from language model activations using

linear probes (Gurnee and Tegmark 2024; Marks and Tegmark 2024). According to the Platonic representation hypothesis (Huh et al. 2024), as training data grows, representations of models trained on different tasks and modalities (vision and language) converge towards a shared statistical model of reality. This could have implications for grounded language models building associations across modalities. However, it does not directly address our main points regarding the mapping metaphor, and thus increasing training data alone is unlikely to lead to genuine symbolic reference.

Another dimension of environment complexity found in more embodied simulations is the inherent environmental dynamics: how does the state and/or changes in the environment affect the agents, and how the agents, in turn, affect the environment? All simulations that incorporate agent movement tap into this dimension. In more dynamic environments, it is easier to release assumptions concerning the descriptive function of communication and meaning-mapping. Naturally, environmental complexity can be exploited only if it aligns with the perceptual and sensorimotor capabilities of the agents.

Finally, the environment may be fully observable—when the agent has full access to the environment's state—or only partially observable. Even embodied simulations in the style of Mordatch and Abbeel (2018) tend to prefer fully observable environments, which are easier for agents to learn in and have less ambiguous communication. The ambiguity introduced by partially observable environments may add to the complexity of communication and help to further relax meaning-mapping assumptions (Gregoromichelaki and Kempson 2019; Rączaszek-Leonardi et al. 2014).

### 5.5 Agents' architecture and learning mechanism

The agents' policies are constrained by the agents' architecture (e.g. feed-forward or recurrent neural network, set of stimuli-response rules, differential equation), among other things. Humans possess certain cognitive biases, one notable example being the limitations of working memory (Baddeley 2003). As established by the No-Free-Lunch Theorem (Wolpert 1996), every learning algorithm has a preference for specific problem types and consequently introduces biases that favour certain communication protocols over others. The strength of these biases may be moderated by the flexibility of the algorithm. In contrast to a simple neural network without a hidden layer, as employed in the work of Smith et al. (2003), which can only represent additively linear influences of object properties on signal components, a multilayer network, used in more recent studies (Ren et al. 2020), can represent a broader range of relationships. The introduction of the transformer neural network architecture (Vaswani et al. 2017) has revolutionised natural language processing based on extensive language corpora; however, this architecture carries its own biases that may not align well with human cognition. Some studies have pointed out that transformers possess limited capabilities in modelling formal languages and generalising relations to out-of-distribution data (Hahn 2020; Wang and Steinert-Threlkeld 2023). Generally, we anticipate that novel network architectures will prove beneficial for learning relationships in more intricate task structures. However, they do not directly address our issues with the three simplifying assumptions.

Another point of concern is the agents' learning mechanism (e.g. supervised learning, reinforcement learning, evolutionary algorithm) and its biological realism. For example, Mordatch and Abbeel (2018) used direct weight sharing between agents and the full knowl-

edge of the dynamics of the environment in their optimisation procedure. They backpropagate loss directly through the full model, making it a case of supervised learning. Such a procedure is very efficient but raises the question whether it does not circumvent some important challenges in language learning that shape language structure (in accordance with the pressure for language learnability (Brighton et al. 2005a, b)). On the other side of the spectrum there are models such as Schulz et al. (2012) and Loula et al. (2010) that used specialised vocabulary learning mechanisms, where only signal associations were updated, and other aspects of agents' activity (i.e., locomotion) were not learnable. Such compartmentalisation of agents' learning may prevent the emergence of more interesting phenomena, where behaviour and signalling forms a single, tightly-coupled system (Grouchy et al. 2016).

### 5.6 Communication function, agent goals and autonomy

Obviously, communication can only evolve if it is useful. Seemingly, there is no single universal function of communication in the natural world. Communication appears in different forms, both when communicating individuals have a common goal and when the goals are different, in the context of cooperation between individuals and competition between them, both as faithful and as deceptive communication (Wacewicz and Żywiczyński 2018). In computational simulations, these aspects are usually modelled as the fitness/cost function. In the basic signalling game, successful communication is the ultimate goal and is rewarded directly and symmetrically. Meanwhile, in Cangelosi (2001) and Floreano et al. (2007), it is only a means to successfully locate food, and the rewards are different for the speaker and the receiver. Most models consider cooperative scenarios, although some explore the conditions under which deceptive communication emerges (e.g. Mitri et al. 2009).

Grouchy et al. (2016) included no explicit cost function at all. Instead, artificial agents were allowed to reproduce whenever two of them met in the same place (analogue of sexual reproduction). Communication arose from evolutionary pressure. We argue that, compared to other types of simulations, agents in this kind of simulation have more autonomy and a better chance of evolving a communication system beyond the expectations of the modeller. This kind of autonomy may help relax the assumptions concerning discrete symbols and the descriptive function of communication.

### 5.7 Social adaptations and social relationships

Somewhat related to the above, pressures on the structure of communication arise from living in a society. Social pressures may come in two varieties: special adaptations of individuals to live in a society or social scaffolding that shapes the interactions between individuals. Adaptations for social behaviour in the form of mechanisms loosely inspired by the Theory of Mind (and following the Rational Speech Act framework (Frank and Goodman 2012)) are included in some models of emergent communication (Batali 1998; Jaques et al. 2019; Bard et al. 2020). These mechanisms use agents' internal cognitive modules to explicitly predict the behaviour of another agent. This basic schema can be studied with only two agents present. While introducing such mechanisms does not lift the meaning-mapping assumption, it makes the mapping truly bi-directional, as signal production includes the consideration of signal interpretation.

Furthermore, social structures, such as social networks (Baronchelli et al. 2006; Zubek et al. 2017), social ecology (Loula et al. 2010), and kinship structures (Floreano et al. 2007; Mitri et al. 2009), have also been demonstrated to be significant in shaping emerging signals. These effects can only be investigated within a larger population comprising diverse agents.[9] It is crucial to consider individual differences among the agents, as these variations introduce ambiguity in communication and differentiate the games the agents play. Particular types of social network topology might make the whole system more susceptible to random fluctuations (Raviv et al. 2020). The experiments by Floreano et al. (2007) suggest that it is evolutionary advantageous to disclose certain pieces of information with related agents while concealing this information from unrelated agents. More intricate social interactions are necessary to incorporate varied functions of communication contingent upon social relations and scaffold the emergence of symbols from continuous actions (Rączaszek-Leonardi 2016; Rączaszek-Leonardi et al. 2018).

### 5.8 Historicity of interactions

As with all complex evolving phenomena, natural languages bear the mark of their long history of evolution. The current shape of the communication system may reflect preadaptations to past evolutionary pressures which are no longer present. At the level of individual language learning, there is a historical trajectory of language development that undergoes different stages. Historicity understood in this way is explored only by a couple of models of emergent communication. The paradigm of iterated learning Kirby et al. (2014), which models vertical language transmission, taps into some aspects of historicity and the information bottleneck associated with it. In Schulz et al. (2012), language games were played in a certain order, and the terms grounded using previous games were used in subsequent games. In other research, Zubek et al. (2017) studied the reactions of a population of agents to a certain trajectory of changes in the environment and demonstrated that the structure of language categories is affected by history. Korbak et al. (2021) presented a developmentally motivated model in which an agent undergoes two stages of learning, which helps it develop a compositional language. We view this form of historicity as necessary for language to move away from a simple description of what is present in the environment and become more abstract, relaxing the assumptions of the descriptive function of communication and meaning-mapping.

## 6 Strongly embodied models combine multiple sources of complexity

As seen in the previous section, there are multiple sources of structure in emerging communication protocols. While some of them can be explored independently to answer specific questions and disentangle relative influences of different factors, we expect that specific properties of symbolic communication may arise as emergent effects of a combination of multiple factors (Rączaszek-Leonardi and Deacon 2018). Thus, models targeting the emergence of symbols should avoid over-simplifications that prevent these factors from interacting, as this in turn may prevent the emergence of certain phenomena. Of course, it is

[9] We can talk about a population only when agents truly differ. This is precluded by models that use direct neural network weight sharing between agents, as in Mordatch and Abbeel (2018).

infeasible and impractical to build a model that is realistic in every possible dimension; models are only useful as abstractions of reality (Ohmer and Cuskley 2023). We observe that a certain strain of embodied simulations naturally combines multiple sources of communication complexity.

First, we must specify what we mean by "embodied simulations". Often, any model or theory that considers an influence of bodily factors on cognition is labelled as "embodied". However, from the perspective of ecological psychology, this is a misconception (Wilson and Golonka 2013). "Embodiment", in a stronger sense, means that cognition is no longer realised by the agent alone but by the relations between the agent and the environment. The environment has to be sufficiently rich and endowed with its own dynamics. Embodied agents' actions are always situated within some ecological context and impact the environment; in turn, the environment provides feedback for the agent. Calls for the ecological realism of the simulations must not be conflated with biological realism. Simulations are typically considered biologically realistic when parts of agents' cognitive architecture are inspired by the brain's anatomy or replicate functions attributed to human cognition (such as the Theory of Mind module). From our perspective, ecological realism consisting of sufficiently complex interactions with the environment is at least equally important. Quoting Mace (2017), "ask not what's in your head, but what your head is inside of". Agents together with their environment form complex dynamic systems and should be modelled as such (Castro 2019).

Simulations have already used simple agents in rich environments, which combine multiple sources of communication complexity and relax at least some of the three assumptions concerning communication emergence modelling identified in Sect. 4. For instance, in Floreano et al.'s (2007), physical space constraints (the inability of multiple robots to occupy the same spot) affected the socio-ecological function of communication. In this work, the robots had to locate and attach themselves to a food source, but the limited space allowed a limited number of robots to feed simultaneously. Thus, communicating the location of food posed the risk that another agent would exploit the scarce resource. This implicit cost of communication, depending on the evolutionary selection mechanisms, resulted in various communication strategies, from honest signalling to deceptive communication (Mitri et al. 2009). In some cases, the robots also learnt to exploit the physical properties of the medium of communication, as they hid emitted light signals from the other robots by hiding behind the food source (Lehman et al. 2018). This is an example of signalling-dependent emergent behaviour, which could not evolve in a simpler environment and demonstrates the complex bidirectional relation between signalling, behaviour, and the environment. The role of signals in these simulations goes beyond simply referring to objects or actions and involves guiding actions in the continuous environment (lifting the meaning-mapping assumption).

The environment adopted by Grouchy et al. (2016) was minimalistic in that it was devoid of any objects apart from the agents. Still, the agents were situated in the sense that they could freely move in a continuous space and use their own locations as a context for communication. Importantly, the emitted signals were continuous and interpreted as such (lifting the assumption about preexisting discrete signs). The continuity of signals opened new possibilities for meaning emergence, as the signal intensity could convey information on the distance in space (basic iconic form of significance). What emerged was not a static signal-referent mapping, which can be used to interpret only previously encountered signals, but a dynamical structure using signals to control agent actions allowing for generalisation.

Ultimately, this led to a dialogical form of communication by which the signals of both agents became entangled. We view the direction of simulating situated dialogue in a dynamical environment as potentially fruitful, as it enables functional communication to emerge with no obvious information asymmetry between the sender and receiver or straightforward descriptive interpretation (Bancerek and Zubek 2023) (lifting the assumption about communication function).

Situatedness is also naturally connected with aspects of perceptual and sensorimotor grounding. It has been shown that situating agents in more complex environments benefits systematic generalisation. Although they did not directly address the problem of language emergence, Hill et al. (2020) presented a model in which they studied the effects of embodiment on language comprehension. Their agent had to navigate through the environment and perform actions according to the instructions given as a sentence in natural language. They demonstrated that (i) egocentric perspective (as opposed to third-person view), (ii) 3D simulated environments (as opposed to 2D grid worlds), and (iii) active perception (as opposed to perceiving the stimuli statically) incentivised the agent to factorise experience and behaviours into pragmatically motivated reusable chunks, leading to better generalisation as shown in experiments isolating factors (i)–(iii).

These results explain previous results showing a low systematic generalisation in recurrent neural networks trained on symbolically encoded stimuli from a single modality (Lake and Baroni 2017). Hill et al. (2020) hypothesised that this effect arises because richer experience is a form of implicit data augmentation. This result suggests that the human capacity to exploit the structure of the world, when learning to generalise in systematic ways, could be replicated in artificial neural networks if such networks are given access to a rich, interactive, multimodal stream of stimuli that adequately matches the experience of an embodied human learner (Bisk et al. 2020).

Strongly embodied models simultaneously integrate multiple sources of communicative complexity and alleviate the three identified assumptions commonly found in communication emergence models. We believe that by pursuing this approach and constructing simulations incorporating non-trivial agent-environment relations, yet maintaining simplicity for the sake of analysis, we can provide a more accurate portrayal of the initial challenges faced by ecological communication systems before they evolve into more abstract and symbolic forms. Strongly embodied simulations facilitate modelling iconic signs, whose form resemble the structure of actions they control, or indexical signs, which are meaningfully embedded within dynamic streams of events (Deacon 1997; Hutchins and Johnson 2009). Incorporating icons and indices as basic classes of signs would open up possibilities for modelling the emergence of symbolic signs following the "ungrounding" path, wherein signs gain symbolic properties gradually as they become less reliant on their immediate context and more dependent on systematic sign-sign relations (Rączaszek-Leonardi and Deacon 2018). Placing an agent within a complex ecology that encompasses not only physical environment interactions but also intricate social dynamics may promote the development of properly symbolic communication systems (Rączaszek-Leonardi et al. 2018).

## 7 Conclusion

Models of communication emergence come from multiple traditions, which explains the differences in terminology and emphasis. The lack of a systematic understanding of the assumptions underlying various models creates a barrier preventing researchers from one domain from using the theoretical insights developed in other domains. We hope that our review will help build a systematic understanding.

As our analysis clarifies, many pressures shape communication signs and symbols, and building a model means bracketing some of them out. Sometimes, however, the bracketing process is too heavy and too hasty, and it may remove the vital elements of the explanation. Perhaps the fascination with one particular kind of language game involving a sender, a receiver, and clearly distinguished referents made us forget that people play many different games (Berne 1964). We propose a change to a dynamic perspective that advocates embodied and situated models, allows the historicity of communication systems to be recognised, and does not presuppose the existence of readily distinguishable symbols. This perspective extends the role of language from describing the world to controlling interactions. We have systematically argued that the mapping metaphor cannot adequately explain the meaning and role of language and should thus not form the foundation of communication emergence models. Instead, models in which meaning emerges through interactions of situated agents in an ecological niche could take on a more central role.

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### Declarations

**Conflict of interest** The authors declare no competing interests.

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